

Ex. Find the non-singular matrices
P and Q such that PAQ is
normal form of

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & 7 \end{bmatrix}$$

Soln: Let $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & 7 \end{bmatrix}_{3 \times 4}$

$\therefore$ we have $A_{3 \times 4} = \underset{\substack{R \\ C}}{I_3} A \underset{C}{I_4}$

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_2 - 2R_1, R_3 + R_1$ Row Column

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 5 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_2 - 2C_1, C_3 + C_1, C_4 - 4C_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 5 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & 1 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_3 \leftrightarrow C_2$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & -3 \\ 0 & 5 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1 & -2 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\frac{1}{5} C_2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 \\ 0 & 1 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1/5 & -2 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 1/5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 - R_2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 14 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1/5 & -2 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 1/5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_4 + 3C_2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 14 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1/5 & -2 & -17/5 \\ 0 & 0 & 1 & 0 \\ 0 & 1/5 & 0 & 3/5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_4 \leftrightarrow C_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1/5 & -17/5 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 1/5 & 3/5 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\frac{1}{14} C_3 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 1/5 & -17/70 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 1/5 & 3/70 & 0 \\ 0 & 0 & 1/14 & 0 \end{bmatrix}$$

$$[I_3 | 0] = PAQ \quad \therefore \rho(A) = 3$$

Where $P = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}$ and

$$Q = \begin{bmatrix} 1 & 1/5 & -17/70 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 1/5 & 3/70 & 0 \\ 0 & 0 & 1/14 & 0 \end{bmatrix}$$